Chapter 21 | Written solutions

Search, Retrieval, and Grounded Assistants

Work through all twelve exercises, from lexical scores and context budgets to evidence support and evaluation design. The examples use fictional office notices. Exercises 9-12 are optional extensions. The written edition keeps every explanation and plot; the notebook PDF also includes code and saved results. The design exercises include illustrative annotations and controlled checks, not measurements of a trained assistant.

1. Compare two query words

The first word appears in only two of ten passages. Matching that word is more distinctive than matching one present in eight passages. Document frequency counts passages containing the word, not all its occurrences. Substitute the two counts into the positive IDF formula:

$$\mathrm{IDF}_2 = \ln\left(1 + \frac{10 - 2 + 0.5}{2 + 0.5}\right) = \ln(4.4) \approx 1.482,$$
$$\mathrm{IDF}_8 = \ln\left(1 + \frac{10 - 8 + 0.5}{8 + 0.5}\right) = \ln(22/17) \approx 0.258.$$

With equal positive frequencies and equal length factors in the candidate, both IDFs are multiplied by the same positive number. The rarer word contributes more. If neither word occurs in the candidate, both contributions are zero regardless of their IDFs.

2. Change the query direction

The query $(0, 1)$ reads the second coordinate. P1, P2, and P3 receive scores 0.6, 0.8, and 1.0, so the descending order is P3, P2, P1. For a query (x, y), subtract the P2 score from the P1 score:

$$(0.8x + 0.6y) - (0.6x + 0.8y) = 0.2(x - y).$$

P1 has a strictly higher score exactly when $x > y$. If $x = y$, the scores tie; the ID tie break still places P1 first, but that ordering is not evidence of a stronger match. The supplied passage vectors have unit length. The function computes raw dot products and does not normalize other inputs.

3. Fit a short context

Visit P1, P2, P3, P4 in order. The rule checks every passage, including those after one that does not fit. At budget 11 the trace is:

Passage	Cost	Before	Decision	After
P1	9	11	Keep	2
P2	5	2	Skip	2
P3	4	2	Skip	2
P4	2	2	Keep	0

The result is P1, P4. At budget 14, P1 leaves five tokens and P2 uses all five; P3 and P4 are skipped. The result is P1, P2. Ranking gives each passage its turn, while its length determines whether it fits at that turn. This greedy procedure does not optimize the number of passages or the coverage of the answer. The notebook uses repeated-word texts only to give each item its prescribed cost.

4. Count missing evidence

The gold evidence set is P2 and P4, so the denominator stays at two. In the ranking P3, P4, P1, P2, P4 arrives second and P2 arrives fourth.

Rank cutoff	Required IDs found	Recall	Any found?	Both found?
1	None	0	No	No
2	P4	0.5	Yes	No
4	P4, P2	1	Yes	Yes

At rank two, a hit-rate check reports success even though half the required evidence is missing. For this question both passages are needed; an any-evidence hit is therefore too weak to establish answerability. If multiple passages were independent alternatives, one could be enough.

5. Mix before or after multiplying

Keep the candidate (y_1, y_2) fixed. Passage A assigns it probability $0.8(0.3) = 0.24$ and B assigns it $0.2(0.9) = 0.18$. RAG-Sequence averages these two path probabilities with the supplied weights:

$$0.6(0.24) + 0.4(0.18) = 0.216.$$

RAG-Token first averages at each position. The chosen first token has probability $0.6(0.8) + 0.4(0.2) = 0.56$. After that same first token, the chosen second token has probability $0.6(0.3) + 0.4(0.9) = 0.54$. Their product is

$$0.56(0.54) = 0.3024.$$

The sequence mixture keeps one passage choice across both positions. The token mixture permits a different latent passage choice at each position. Mixing and multiplying therefore need not commute. Neither computation issues a new search for the second token; both use the original retrieved set. These are probabilities of the two-token prefix; a complete answer also includes its end marker. The larger probability here does not establish better answer quality.

6. Turn off length adjustment

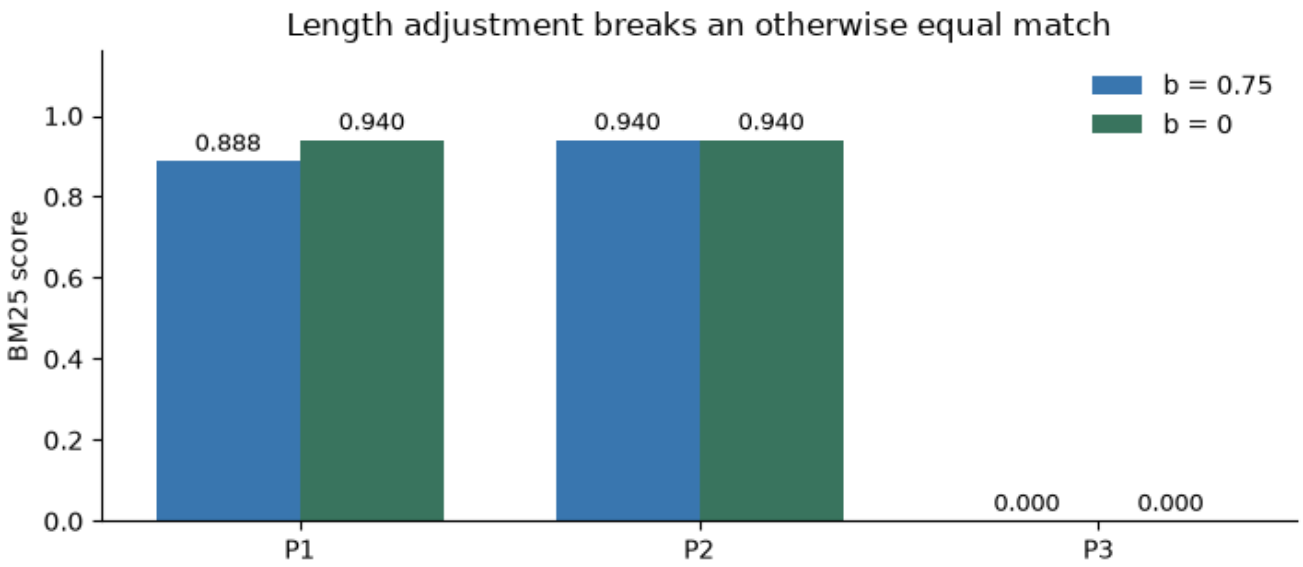
P1 tells a resident where to renew a parking permit. P2 gives the expiry date. Both notices contain *parking* and *permit* once, so the two notices are a good test of how BM25 uses passage length. Score the three notices from Table 20.1 in ID order for `parking permit`. Then change only the length parameter `b` from its default value of `0.75` to `0`.

The passages have 8, 7, and 6 word tokens, so the collection average is 7. Each query word occurs in two of the three passages and contributes the same inverse document frequency, $\ln(1.6) \approx 0.470004$. In P1 and P2, each query word occurs once. With the chapter's default $k_1 = 1.2$, the score for either matching passage of length L is therefore

$$2\ln(1.6)\frac{2.2}{1 + 1.2(1 - b + bL/7)}.$$

With $b = 0.75$, the denominator is about 2.329 for P1 and 2.2 for P2. The smaller denominator gives the shorter notice, P2, the higher score. P3 contains neither query word and scores zero in both runs.

With $b = 0$, both denominators are 2.2 and both scores equal $2\ln(1.6)$. `rank_passages` orders equal scores by ascending passage ID, so P1 comes first. The rank after the tie break carries no extra evidence about which passage best answers the question.



The plot isolates passage length: both matching notices contain the same query terms once, but P2 is shorter. The displayed scores are ranking quantities, not probabilities of answer correctness.

7. Rerank only what was found

For "Where do I renew a parking permit?" the only passage that supplies the location is P1. Compare a retrieval run that returned P2 and P3 with one that also returned P1. Keep P2 and P3's scores fixed so we can see what changes when P1 becomes available.

Recall at rank one is the number of required evidence IDs in the first result divided by the total number of required IDs. Here there is one required ID, P1.

With **P2 and P3**, P3 comes first because $0.7 > 0.2$. The top result contains zero of the one required IDs, giving recall $0/1 = 0$.

With **P2, P3, and P1**, the order is P1, P3, P2 because $0.9 > 0.7 > 0.2$. The top result contains P1, giving recall $1/1 = 1$.

Retrieval chooses which passages the reranker can consider. Reranking changes their order using the supplied query–passage scores. When retrieval includes P1, its score moves it from third to first. When retrieval omits P1, changing the order of P2 and P3 leaves the evidence absent.

8. Keep the passage and ID together

The answer to "Where can I renew a parking permit?" is in P1. Begin with the ranking P2, P1, P3 and see whether P1 makes it into the context. Then put P1 first and repeat the comparison at the same budget of 14 tokens.

A passage's cost is its number of tokens under the chapter's teaching tokenizer. It lowercases words, ignores the final periods, and counts the number 30 as one token. The budget here covers only passage text. The ID travels with the text as a separate field.

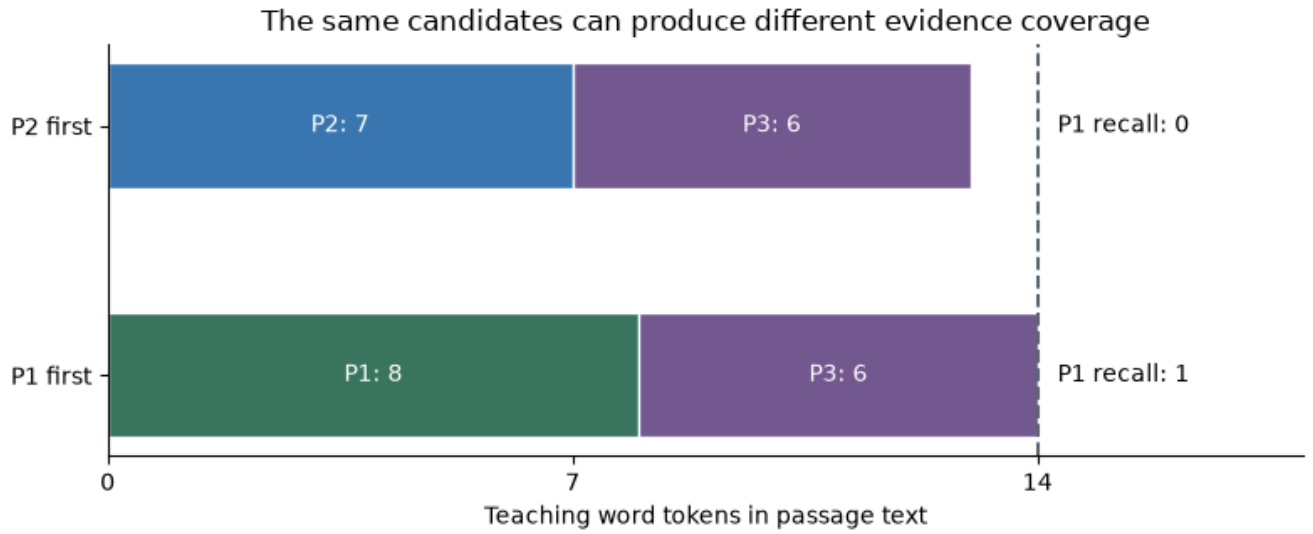
`pack_context` reads the ranking from left to right. It keeps a whole passage if its cost is at most the remaining budget. If a passage is too long, it skips that passage and tries the next one. Each retained pair includes the original passage ID and the full text from the dictionary.

P2, P1, P3 with budget 7. P2 uses all seven tokens. Neither P1 nor P3 fits in the zero tokens left. The packed context contains only P2, and the phrase has no hits.

P2, P1, P3 with budget 14. P2 leaves seven tokens. P1 needs eight, so it is skipped. P3 needs six, so it fits and leaves one token. The packed context contains P2 and P3, and the phrase has no hits.

P1, P2, P3 with budget 14. P1 leaves six tokens. P2 needs seven, so it is skipped. P3 fits exactly. The packed context contains P1 and P3, and the hit is P1.

The same three passages were retrieved in every run. At budget 14, the first of P1 and P2 to be visited uses enough of the budget to exclude the other, since their combined cost is 15. The first two runs lose the renewal location during packing. Keeping each ID attached to its text lets us check which passage actually reached the input and what a citation to that passage would support.



Both bars satisfy the budget of 14. The dashed line marks the budget, and the first bar leaves one token unused. Retrieval found P1 in both cases; only visiting P1 first preserves it in the final context. The reported recall uses the packed IDs, with the same one-passage gold set.

9. Probe a literal match (optional)

The claim we want to check is “Express renewals are available.” Search the two notices for the shorter phrase `express renewals`. The function lowercases both the phrase and the passage text, then checks whether the phrase’s tokens occur together in order within each passage. The rest of the sentence does not enter that check.

Both notices begin with the consecutive tokens `express` and `renewals`, so both IDs match. P4 states that express renewals are available. In P5, “no longer” changes the claim about current availability. P5 contradicts the positive claim.

A phrase hit can help locate the passage to read. To judge support for the positive claim, read what each passage says about the service, including any negation. A notice that says “You can renew through the express service” could support the same claim even though this phrase search would miss it.

10. Choose a passage boundary (optional)

Consider: "Renew a parking permit online. If your address changed this month, renew in person at the mobility desk." A resident who moved this month needs the exception. Answering with the general online rule would misapply the notice even though those words really occur in the source.

Compare two indexes. One stores the sentences separately under distinct IDs with their parent document and offsets; the other stores the whole notice. Keep P2 and P3 as distractors. Splitting can return the general rule without its exception. Keeping the notice together preserves the relationship when the entire passage fits, but the longer passage may be excluded by a small budget. Neither boundary choice always wins.

Before running retrieval, label supporting spans in the original document. Use those spans as the scoring units in both indexes. Counting passage IDs would change the denominator when two passages become one. Include questions about the ordinary route, a recent move, and both routes together; add paraphrases such as "I relocated this month. How do I extend my parking authorization?" Keep queries, retriever, rank cutoff, packing policy, token budget, and generator settings fixed, rebuilding each index as needed.

Record required-span coverage at retrieval and after packing. For a question asking both routes, require both spans. Repeat at a larger budget to separate boundary effects from space limits. Score the answer independently: the recent-move case must direct the resident to the mobility desk and preserve the condition; the both-routes case needs the ordinary rule as well. Check citations against the actual supplied text. A control that supplies both sentences directly tests whether the generator can use evidence when search and packing have already succeeded.

The notebook fixture records which annotated spans are visible. It demonstrates the scoring unit; it does not run a retriever or generator.

11. Design an abstention test (optional)

Use the three notices as the entire collection. An abstention explicitly says that this collection supplies insufficient evidence for the question.

Question	Expected behavior
When does a parking permit expire?	Answer June 30 and cite P2.
Is there a grace period after expiry?	Say the notices do not specify one.
Who won the district chess tournament?	Say the notices do not provide this information.

The grace-period question shares vocabulary with P2. Quoting its expiry date does not answer the question. Saying "There is no grace period" would also go beyond the evidence: absence from the collection leaves its status unknown. A supported fact can explain an abstention, but cannot replace it.

For the answerable case, score the correct date, a supporting citation to P2 from the actual prompt, and absence of invented additions separately. Count a refusal as a missed answer in the end-to-end task. For each unanswerable case, credit an explicit statement of insufficient evidence without an invented answer. Changing the subject earns no credit.

For a larger test, report grounded-answer rate and false-abstention rate on answerable questions. Report appropriate abstentions and unsupported answers separately on related and unrelated unanswerable questions. Use the number of questions in each group as its denominator and show counts. In this three-case fixture, always abstaining passes two cases overall but answers zero of the one answerable questions.

Log retrieved and packed IDs. If P2 never reached the prompt, refusing the expiry question can be appropriate for the generator while the full system still failed to answer an answerable question. Supplying P2 directly is a useful control. Empty gold evidence yields undefined retrieval recall; it should not silently become zero. Here unanswerability was established by reading all three notices, not inferred from missing labels.

The notebook evaluates an explicitly supplied always-abstain fixture. It is a rubric sanity check, not an observed model benchmark.

12. Audit two versions (optional)

Keep P1 as the current notice directing renewals to the mobility desk. Add P1-old directing them to the records counter. Both belong to the same office and policy. A query containing "renew a parking permit" matches both; the applicable date determines which text governs the answer.

Record stable version-specific passage IDs, parent document and offsets, issuing office or jurisdiction, version identifier, source location, effective start and end dates, and explicit supersession links. Keep publication, upload, and effective dates distinct. A later upload need not be the currently applicable rule. Preserve old versions so a citation remains traceable and historical questions can be answered.

For a fictional test, let P1-old apply from January 1 through March 31, 2026, and P1 from April 1 onward. On April 15 the expected location is mobility desk with P1; on March 15 it is records counter with P1-old. Define the clock for undated questions, for example April 15, and save the collection snapshot. Add paraphrases and queries sharing the exact renewal words. Label the applicable ID and location before running.

Compare lexical-only retrieval with a version-aware run under the same rank cutoff and budget. One controlled policy filters by office and effective interval before ranking eligible notices. These two texts both have eight teaching tokens and contain each query term once, so their BM25 scores tie. P1 winning the ID tie break does not demonstrate use of dates. Historical queries expose that shortcut.

Record applicable-ID recall among candidates, survival after packing, answer location, citation support, and temporal applicability separately. A mobility-desk answer citing P1-old has the right current location but an unsupported citation. A records-counter answer citing P1-old may quote its source faithfully while using the wrong version for a current question. If both notices are shown, the answer must resolve their dates rather than present both desks as current choices. Report current and historical results separately. If authority or effective dates leave a genuine conflict, report the uncertainty; choosing the newest timestamp is not a resolution rule.

The notebook check uses half-open intervals: the start is included and the end is excluded. An end of April 1 therefore means validity through March 31. The fixture verifies applicability labels, not model behavior.